```
# Prathamesh Lavange B 31
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
from google.colab import files

uploaded = files.upload()

print("Uploaded files:", list(uploaded.keys()))
```

```
Choose Files No file chosen  Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving sales.json to sales.json
Saving sales.xlsx to sales.xlsx
Saving sales.csv to sales.csv
Uploaded files: ['sales.json', 'sales.xlsx', 'sales.csv']
```

```
df_csv = pd.read_csv("sales.csv")
df_excel = pd.read_excel("sales.xlsx")
df_json = pd.read_json("sales.json")

print("CSV Data:")
display(df_csv.head())

print("Excel Data:")
display(df_excel.head())

print("JSON Data:")
display(df_json.head())
```

```
CSV Data:
```

	Order_ID	Order_Date	Customer	Category	Product	Quantity	Price	Region	Total_Sales
0	CA-001	2024-01-05	Aarav	Technology	Laptop	2	55000	West	110000
1	CA-002	2024-01-12	Priya	Furniture	Chair	1	8500	South	8500
2	CA-003	2024-02-03	Rahul	Office Supplies	Paper	10	300	East	3000
3	CA-004	2024-02-18	Sneha	Technology	Mobile	2	25000	West	50000
4	CA-005	2024-03-10	Aditya	Furniture	Table	1	15000	North	15000

```
Excel Data:
```

	Order_ID	Order_Date	Customer	Category	Product	Quantity	Price	Region	Total_Sales
0	CA-001	2024-01-05	Aarav	Technology	Laptop	2	55000	West	110000
1	CA-002	2024-01-12	Priya	Furniture	Chair	1	8500	South	8500
2	CA-003	2024-02-03	Rahul	Office Supplies	Paper	10	300	East	3000
3	CA-004	2024-02-18	Sneha	Technology	Mobile	2	25000	West	50000
4	CA-005	2024-03-10	Aditya	Furniture	Table	1	15000	North	15000

```
JSON Data:
```

	Order_ID	Order_Date	Customer	Category	Product	Quantity	Price	Region	Total_Sales
0	CA-001	2024-01-05	Aarav	Technology	Laptop	2	55000	West	110000
1	CA-002	2024-01-12	Priya	Furniture	Chair	1	8500	South	8500
2	CA-003	2024-02-03	Rahul	Office Supplies	Paper	10	300	East	3000
3	CA-004	2024-02-18	Sneha	Technology	Mobile	2	25000	West	50000
4	CA-005	2024-03-10	Aditya	Furniture	Table	1	15000	North	15000

```
print("CSV Shape:", df_csv.shape)
print("Excel Shape:", df_excel.shape)
print("JSON Shape:", df_json.shape)

print("\nColumn Names:")
print(df_csv.columns.tolist())
```

```
print("\nDataset Information:")
df_csv.info()
```

```
CSV Shape: (10, 9)
Excel Shape: (10, 9)
JSON Shape: (10, 9)

Column Names:
['Order_ID', 'Order_Date', 'Customer', 'Category', 'Product', 'Quantity', 'Price', 'Region', 'Total_Sales']

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 9 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Order_ID     10 non-null     object
 1   Order_Date   10 non-null     object
 2   Customer     10 non-null     object
 3   Category     10 non-null     object
 4   Product      10 non-null     object
 5   Quantity     10 non-null     int64
 6   Price        10 non-null     int64
 7   Region       10 non-null     object
 8   Total_Sales  10 non-null     int64
dtypes: int64(3), object(6)
memory usage: 852.0+ bytes
```

```
print("Missing Values:")
print(df_csv.isnull().sum())

print("\nDuplicate Rows:", df_csv.duplicated().sum())
```

```
Missing Values:
Order_ID       0
Order_Date     0
Customer       0
Category       0
Product        0
Quantity       0
Price          0
Region         0
Total_Sales    0
dtype: int64

Duplicate Rows: 0
```

```
df = df_csv.copy()

df = df.drop_duplicates()

df["Order_Date"] = pd.to_datetime(
    df["Order_Date"], errors="coerce"
)

df["Quantity"] = pd.to_numeric(
    df["Quantity"], errors="coerce"
)

df["Price"] = pd.to_numeric(
    df["Price"], errors="coerce"
)

df["Total_Sales"] = pd.to_numeric(
    df["Total_Sales"], errors="coerce"
)

df = df.dropna(
    subset=["Order_Date", "Quantity", "Price", "Total_Sales"]
)

print("Data cleaning completed!")
display(df.head())
```

```
Data cleaning completed!
```

	Order_ID	Order_Date	Customer	Category	Product	Quantity	Price	Region	Total_Sales
0	CA-001	2024-01-05	Aarav	Technology	Laptop	2	55000	West	110000
1	CA-002	2024-01-12	Priya	Furniture	Chair	1	8500	South	8500
2	CA-003	2024-02-03	Rahul	Office Supplies	Paper	10	300	East	3000
3	CA-004	2024-02-18	Sneha	Technology	Mobile	2	25000	West	50000
4	CA-005	2024-03-10	Aditya	Furniture	Table	1	15000	North	15000

```
df["Calculated_Sales"] = df["Quantity"] * df["Price"]

print("Total Sales:", df["Total_Sales"].sum())
print("Calculated Sales:", df["Calculated_Sales"].sum())

display(df[[
    "Order_ID", "Product", "Quantity",
    "Price", "Total_Sales", "Calculated_Sales"
]])
```

```
Total Sales: 235750
Calculated Sales: 235750
```

	Order_ID	Product	Quantity	Price	Total_Sales	Calculated_Sales
0	CA-001	Laptop	2	55000	110000	110000
1	CA-002	Chair	1	8500	8500	8500
2	CA-003	Paper	10	300	3000	3000
3	CA-004	Mobile	2	25000	50000	50000
4	CA-005	Table	1	15000	15000	15000
5	CA-006	Binder	8	250	2000	2000
6	CA-007	Monitor	2	12000	24000	24000
7	CA-008	Desk	1	18000	18000	18000
8	CA-009	Pen	15	50	750	750
9	CA-010	Keyboard	3	1500	4500	4500

```
print("Statistical Summary:")
display(df[["Quantity", "Price", "Total_Sales"]].describe())

print("Average Sales:", df["Total_Sales"].mean())
print("Maximum Sales:", df["Total_Sales"].max())
print("Minimum Sales:", df["Total_Sales"].min())
```

```
Statistical Summary:
```

	Quantity	Price	Total_Sales
count	10.000000	10.000000	10.000000
mean	4.500000	13560.000000	23575.000000
std	4.836206	16927.686066	33785.691005
min	1.000000	50.000000	750.000000
25%	1.250000	600.000000	3375.000000
50%	2.000000	10250.000000	11750.000000
75%	6.750000	17250.000000	22500.000000
max	15.000000	55000.000000	110000.000000

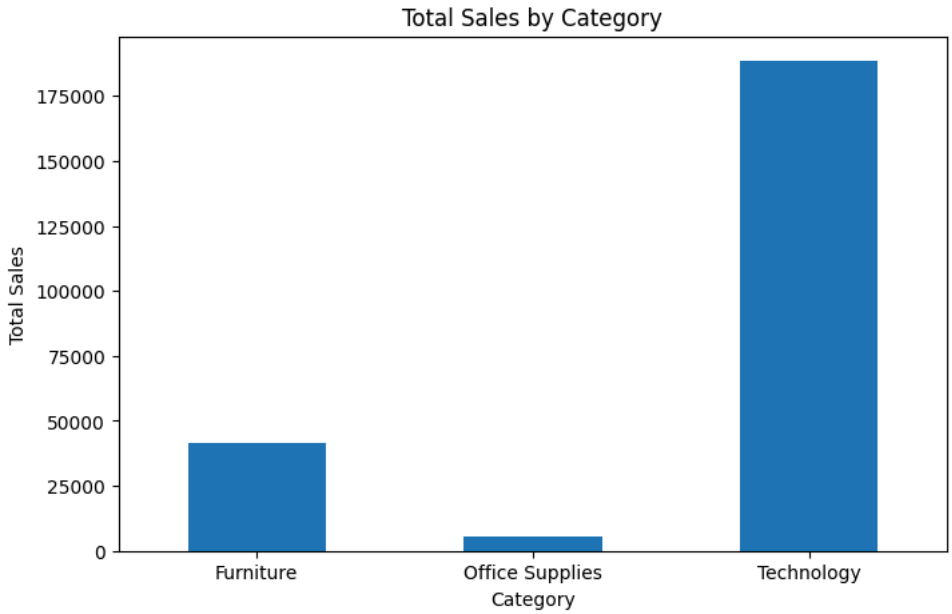
```
Average Sales: 23575.0
Maximum Sales: 110000
Minimum Sales: 750
```

```
category_sales = df.groupby("Category")["Total_Sales"].sum()

print(category_sales)

category_sales.plot(kind="bar", figsize=(8, 5))
plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.show()
```

```
Category
Furniture          41500
Office Supplies     5750
Technology        188500
Name: Total_Sales, dtype: int64
```

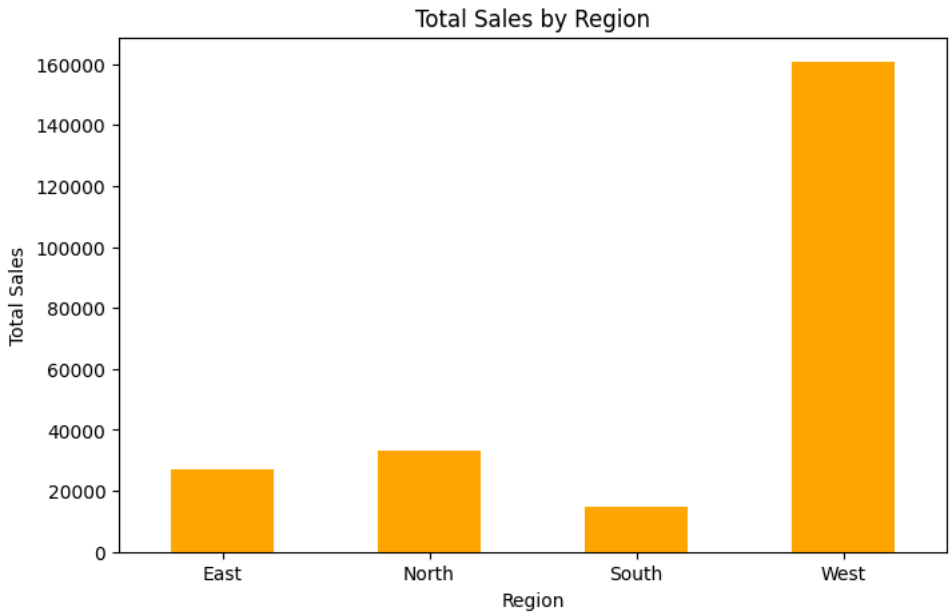



```
region_sales = df.groupby("Region")["Total_Sales"].sum()

print(region_sales)

region_sales.plot(kind="bar", color="orange", figsize=(8, 5))
plt.title("Total Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.show()
```

```
Region
East      27000
North     33000
South     15000
West     160750
Name: Total_Sales, dtype: int64
```



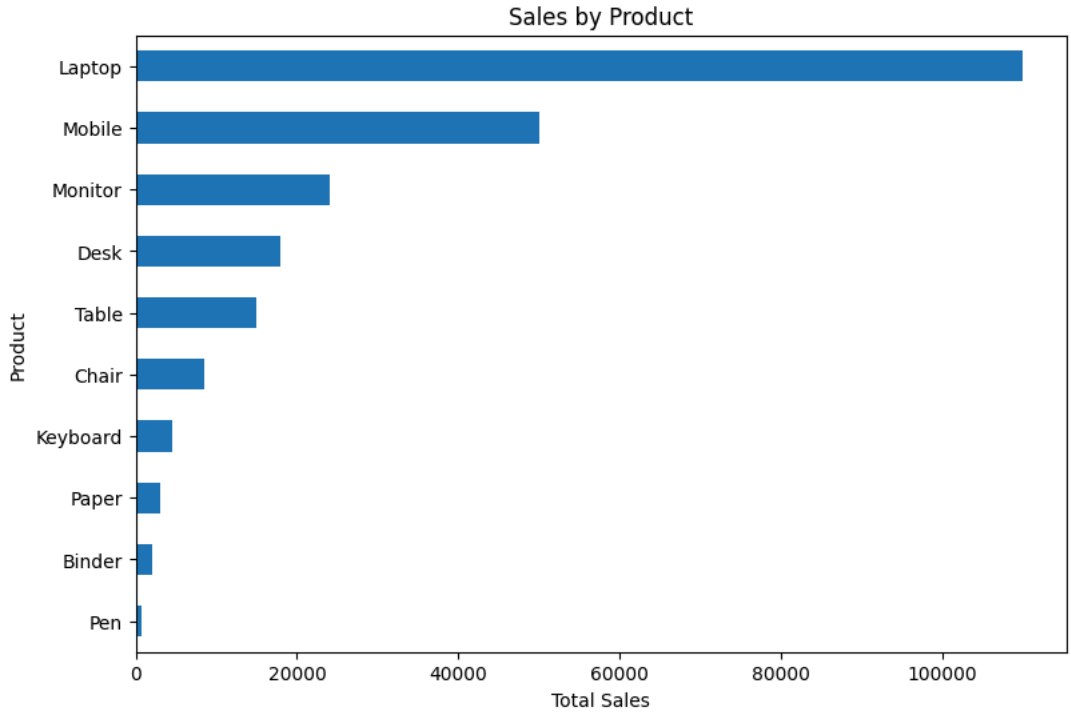

```
product_sales = df.groupby("Product")["Total_Sales"].sum()

print(product_sales.sort_values(ascending=False))

product_sales.sort_values().plot(
    kind="barh", figsize=(9, 6)
)
plt.title("Sales by Product")
```

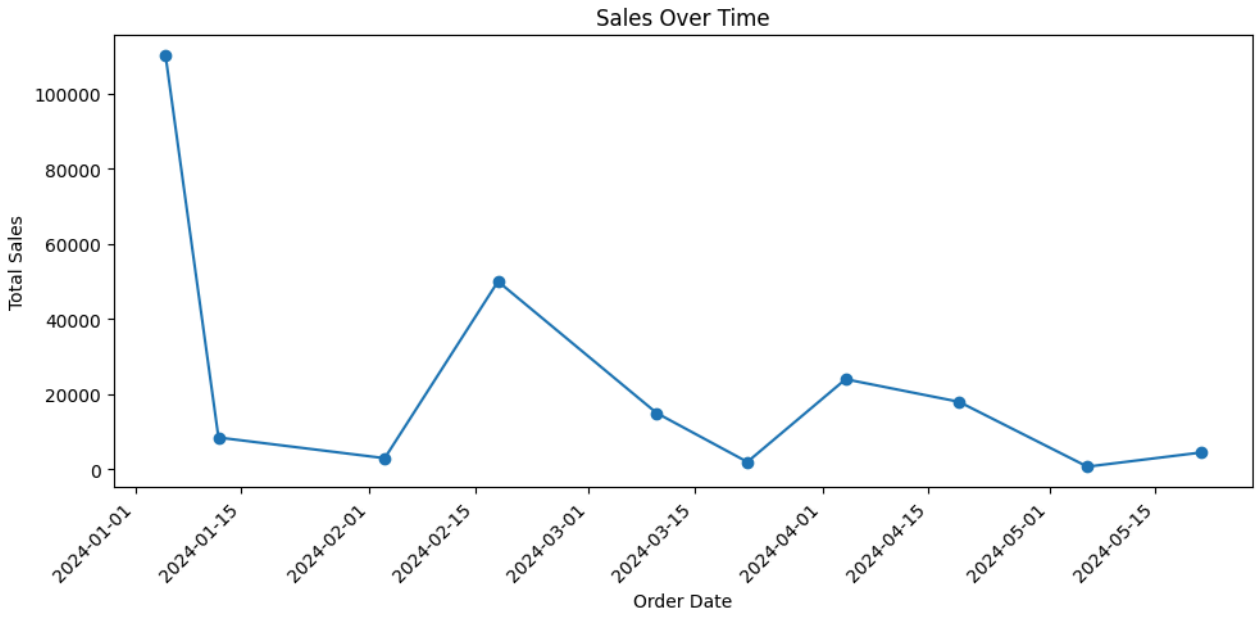
```
plt.xlabel("Total Sales")
plt.show()
```

```
Product
Laptop      110000
Mobile       50000
Monitor      24000
Desk         18000
Table        15000
Chair         8500
Keyboard      4500
Paper         3000
Binder        2000
Pen            750
Name: Total_Sales, dtype: int64
```




```
daily_sales = df.groupby("Order_Date")["Total_Sales"].sum()

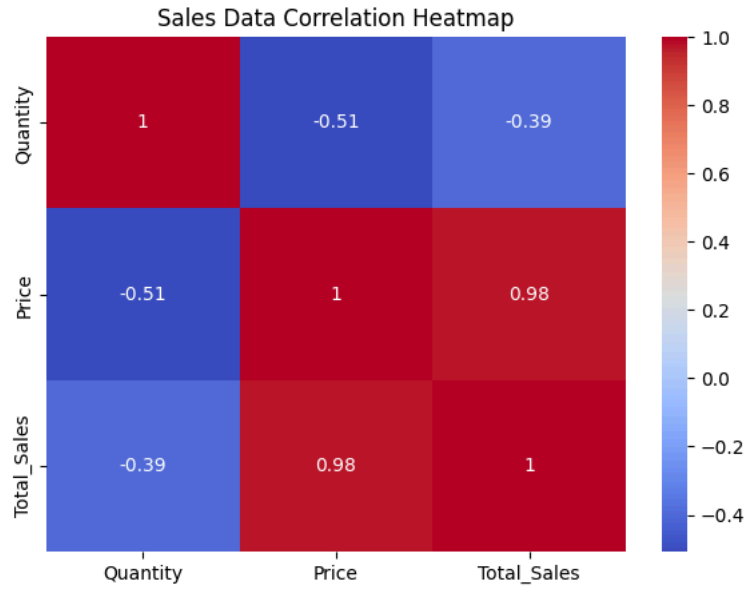
daily_sales.plot(marker="o", figsize=(10, 5))
plt.title("Sales Over Time")
plt.xlabel("Order Date")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
plt.figure(figsize=(7, 5))

sns.heatmap(
    df[["Quantity", "Price", "Total_Sales"]].corr(),
    annot=True,
    cmap="coolwarm"
)

plt.title("Sales Data Correlation Heatmap")
plt.show()
```




```
df.to_csv("cleaned_sales.csv", index=False)
df.to_excel("cleaned_sales.xlsx", index=False)
df.to_json("cleaned_sales.json", orient="records", indent=4)

print("All cleaned files saved successfully!")
```

```
All cleaned files saved successfully!
```

```
from google.colab import files

files.download("cleaned_sales.csv")
files.download("cleaned_sales.xlsx")
files.download("cleaned_sales.json")
```